

Graph-signal-processing foundations for cycling-network expansion: spectral bounds, optimal siting and empirical learning curves on dock-based bike-share systems

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Abstract

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Dock-based bike-share operators routinely face the same question : where should the next K stations go on a network of N already-deployed sites ? Two literatures answer separately and inconclusively : operations-research coverage models (MCLP, k -median) ignore the spatial auto-correlation of demand, and prediction-based feature models require historical trip data that candidate sites by definition do not have.

We close this gap with a Graph-Signal-Processing (GSP) treatment on the station-proximity graph that the two literatures share. Both demand and the supply-side IMD-4 features of the companion paper (Fossé and Pallares, 2026b) live as graph signals on the same Laplacian. Three operational tools follow from a single eigendecomposition :

(I) A cold-start spectral diagnostic. An upper bound on the linear R^2 recoverable from any d -feature combination (Theorem 14) returns $R_{\text{spectral}}^2 \in [0.08, 0.32]$ on nine LSO networks ≥ 400 stations, in agreement with the actual ceiling of non-linear gradient-boosted regression.

(II) A submodular-guaranteed siting algorithm. D -optimal greedy subset selection on the Laplacian eigenbasis, backed by a Nemhauser-Wolfe-Fisher $(1 - 1/e)$ bound, recovers a held-out-station ranking with Spearman $\rho \in [+0.71, +0.77]$ on Boston and DC, dominating the MCLP and k -median baselines at $K = 50$ by $\Delta\rho > +0.6$ on the same panels.

(III) A sample-size saturation diagnostic. The empirical learning curve $\rho(N) = \rho_\infty(1 - \exp(-N/n_*))$ saturates on Boston at $n_* = 21 \pm 3$ training stations, $\rho_\infty = 0.77$, providing an *a priori* criterion for when a partly-deployed network carries enough information for IMD-4 leave-station-out transfer.

The three tools are released as a single Python toolkit and share the data infrastructure of the predictive companion paper (Fossé and Pallares, 2026a).

Scope and contribution boundary. Five of the seven theorems (T1, T2, T3, T4, T5) apply established graph-signal-processing or operations-research results to the bike-share setting – specifically Dirichlet- energy identities, orthogonal-projection R^2 , Nemhauser-Wolfe-Fisher submodularity, and the Anis-Gadde-Ortega band-limited reconstruction bound. Their contribution is one of systematic application. The novel empirical content is the D -optimal vs MCLP / k -median benchmark (Tool II) and the saturation curve fit (Tool III).

Keywords: graph signal processing ; bike-sharing ; station siting ; submodular optimisation ; spectral bounds ; sampling theory ; learning curves.

Highlights

- Spectral upper bound on the linear R^2 recoverable from d supply-side features

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- Submodularity-based $1 - 1/e$ guarantee for D -optimal greedy station siting
- Band-limited reconstruction bound (Anis-Gadde-Ortega) applied to bike-share demand
- Empirical learning curve on Boston: $n_* = 21 \pm 3$ training stations to saturate
- D -optimal siting beats MCLP / k -median by $\Delta\rho > +0.6$ at $K = 50$

1 Introduction

Dock-based bike-share systems pose a recurring operational question : where, on a city map already partially equipped with N stations, should the K next stations go ? Two literatures answer this question separately and inconclusively.

The first literature treats it as a *coverage* problem (Maximal Covering Location, MCLP, Church and ReVelle, 1974; k -median, Hakimi, 1964; PHCS, Lin and Yang, 2013), maximising population or demand covered within a fixed walking radius. Coverage objectives are robust to data scarcity but ignore the spatial auto-correlation of demand : two stations 200 m apart in a business district are treated as independently informative samples, when they are nearly substitutable in their predictive content.

The second literature treats it as a *prediction* problem (gradient-boosted regression on station-level features (Faghih-Imani et al., 2014; Fossé and Pallares, 2026a); geographically weighted regression (Mooney et al., 2019); graph neural networks (Kipf and Welling, 2017; Lin et al., 2018)), targeting the demand a candidate station would realise once deployed. Prediction-based approaches honour spatial auto-correlation but require historical trip data that, by definition, the candidate sites do not have.

We argue that the gap between the two literatures is closed by *Graph Signal Processing* (GSP, Shuman et al., 2013; Sandryhaila and Moura, 2014) on the station- proximity graph. Both demand and supply-side features (the IMD-4 axes of Fossé and Pallares, 2026b) are graph signals on the same object. A spectral analysis of this object yields three operational tools that this paper develops and empirically benchmarks :

Tool 1 – A spectral diagnostic for cold-start prediction. Theorem 14 bounds the linear R^2 recoverable from any d -dimensional feature set against the band-limited content of demand on the Laplacian eigenbasis. Applied to the IMD-4 supply-side features on nine LSO networks (Table 1), the bound returns $R^2_{\text{spectral}} \in [0.08, 0.32]$ – the linear ceiling that a non-linear gradient-boosted regressor recovers up to ~ 0.18 out-of-sample on the same axes (script d36 of the companion repository).

Tool 2 – Sampling-theoretic siting with a submodular guarantee. We cast new-station siting as D -optimal subset selection on the Laplacian eigenbasis (Section 3.6). The Nemhauser-Wolfe-Fisher ($1 - 1/e$) submodular bound (Theorem 4) certifies a greedy procedure ; on Boston and DC the resulting ranking dominates the MCLP / k -median baselines by $\Delta\rho > +0.6$ at $K = 50$ (Table 3). This is the central novel empirical result of this paper.

Tool 3 – A demand-saturation diagnostic for partial deployments. An empirical learning curve on Boston Bluebikes (Theorem 6) saturates at $n_* \approx 21 \pm 3$ training stations, $\rho_\infty = 0.77$. Below the saturation point, the leave-station-out IMD-4 ranking is operationally unreliable ; above it, the IMD-4 carries transferable spatial information that fixed-effect models cannot recover (Fossé and Pallares, 2026b, Section LSO).

The three tools share the same graph object and a single eigendecomposition. Two-thirds of the empirical analyses in this paper inherit their data layer from Fossé and Pallares (2026a) (the same nine-network LSO panel), and the IMD-4 features used throughout are defined and

validated in Fossé and Pallares (2026b). Sections 3.5 (infill smoother), 3.6 (siting algorithm) and 3.7 (robustness checks) develop each tool in turn.

Status note. This is a v0.2 working draft. Five of the seven theorems (Theorems 1, 2, 3, 4, 5) are applications of established results in graph signal processing and operations research – specifically Dirichlet-energy identities, orthogonal-projection R^2 , the Nemhauser-Wolfe-Fisher submodular bound, and the Anis-Gadde-Ortega band-limited reconstruction guarantee. Their contribution to the bike-share use-case is one of *systematic application* rather than new mathematics. Theorems 6 (saturation curve) and 7 (MCLP comparison) are empirical findings stated as testable predictions. The empirical benchmarks of Tool 2 are, to the best of our knowledge, the first systematic comparison of D -optimal greedy siting against MCLP / k -median on a dock-based bike-share panel.

Relation to the companion paper. All empirical results of this paper that involve trip-level data (Sections 3.5, 3.6, 3.7) re-use the data infrastructure of the companion paper (Tier 1 / Tier 2 panels of nine LSO networks). The IMD-4 indicator itself is defined and validated in the companion paper and used here as a fixed supply-side input.

2 Mathematical foundations

This section formalises the graph-signal-processing objects that the rest of the paper rests on, and proves the four results that underpin the empirical claims of Section 3. Theorems 1 and 3 re-state and rigorously prove the two spectral bounds that the §8 extract treats informally; Theorem 5 is a new permutation test for the spectral-smoothness statistic with an explicit non-asymptotic concentration bound; Theorem 7 is a new finite-sample bound relating the linear projection R^2 to the k -fold cross-validation R^2 achievable by any non-linear regressor on the same subspace.

2.1 Setup

Throughout this section we work on a finite, simple, undirected weighted graph $G = (V, E, \mathbf{W})$ with $|V| = N$, $\mathbf{W} \in \mathbb{R}_{\geq 0}^{N \times N}$ symmetric and zero on the diagonal. The (combinatorial) graph Laplacian is $\mathbf{L} = \mathbf{D} - \mathbf{W}$ with $\mathbf{D} = \text{diag}(\mathbf{W}\mathbf{1})$; the symmetric-normalised Laplacian is $\mathbf{L}_{\text{sym}} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{W}\mathbf{D}^{-1/2}$ when $\mathbf{D}$ is invertible. Both are real symmetric positive-semidefinite, and we write their eigendecompositions as $\mathbf{L} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$ and $\mathbf{L}_{\text{sym}} = \mathbf{U}_s\mathbf{\Lambda}_s\mathbf{U}_s^\top$, eigenvalues sorted in non-decreasing order.

A graph signal is a vector $\mathbf{f} \in \mathbb{R}^N$. Its *combinatorial Dirichlet energy* is

$$\mathcal{E}_G(\mathbf{f}) := \mathbf{f}^\top \mathbf{L} \mathbf{f} = \frac{1}{2} \sum_{(i,j) \in E} \mathbf{W}_{ij} (f_i - f_j)^2. \quad (1)$$

The normalised variant $\mathcal{E}_G^s(\mathbf{f}) := \mathbf{f}^\top \mathbf{L}_{\text{sym}} \mathbf{f}$ will appear in Theorem 5.

2.2 Theorem 1 — Permutation expectation of Dirichlet energy

Theorem 1 (Permutation expectation). *Let $\mathbf{f} \in \mathbb{R}^N$ with $\mathbf{1}^\top \mathbf{f} = 0$ and $\|\mathbf{f}\|_2^2 = N - 1$ (i.e. centred with unit sample variance), and let $\Pi \in \mathbb{R}^{N \times N}$ be a uniformly random permutation matrix. Then*

$$\mathbb{E}_\Pi[\mathcal{E}_G(\Pi\mathbf{f})] = \text{tr}(\mathbf{L}) = 2\|E\|, \quad (2)$$

where $\|E\| := \frac{1}{2} \sum_{(i,j)} \mathbf{W}_{ij}$ is the total edge weight. The right-hand side does not depend on $\mathbf{f}$.

Proof. Write $\Pi\mathbf{f} = (f_{\pi(1)}, \dots, f_{\pi(N)})^\top$ for a uniform permutation π . Expanding the quadratic form,

$$\mathcal{E}_G(\Pi\mathbf{f}) = \sum_{i,j} \mathbf{L}_{ij} f_{\pi(i)} f_{\pi(j)}.$$

The expectation $\mathbb{E}_\pi[f_{\pi(i)}f_{\pi(j)}]$ depends only on whether $i = j$ or not. Using $\sum_k f_k = 0$ and $\sum_k f_k^2 = N - 1$,

$$\begin{aligned}\mathbb{E}_\pi[f_{\pi(i)}^2] &= \frac{1}{N} \sum_k f_k^2 = \frac{N-1}{N}, \\ \mathbb{E}_\pi[f_{\pi(i)}f_{\pi(j)}] &= \frac{1}{N(N-1)} \left(\left(\sum_k f_k \right)^2 - \sum_k f_k^2 \right) = -\frac{1}{N} \quad (i \neq j).\end{aligned}$$

Combining,

$$\begin{aligned}\mathbb{E}_\Pi[\mathcal{E}_G(\Pi\mathbf{f})] &= \frac{N-1}{N} \text{tr}(\mathbf{L}) - \frac{1}{N} \sum_{i \neq j} \mathbf{L}_{ij} \\ &= \frac{N-1}{N} \text{tr}(\mathbf{L}) - \frac{1}{N} (\mathbf{1}^\top \mathbf{L} \mathbf{1} - \text{tr}(\mathbf{L})).\end{aligned}$$

For the combinatorial Laplacian, $\mathbf{L}\mathbf{1} = \mathbf{0}$ (the all-ones vector is in the kernel of $\mathbf{L}$), so $\mathbf{1}^\top \mathbf{L} \mathbf{1} = 0$ and the second term collapses to $+\frac{1}{N} \text{tr}(\mathbf{L})$. Adding,

$$\mathbb{E}_\Pi[\mathcal{E}_G(\Pi\mathbf{f})] = \frac{N-1}{N} \text{tr}(\mathbf{L}) + \frac{1}{N} \text{tr}(\mathbf{L}) = \text{tr}(\mathbf{L}).$$

Finally, $\text{tr}(\mathbf{L}) = \sum_i \mathbf{D}_{ii} - \sum_i \mathbf{W}_{ii} = \sum_i \sum_j \mathbf{W}_{ij} = 2\|E\|$ since $\mathbf{W}$ is symmetric with zero diagonal. $\square$

Remark 2. For the symmetric-normalised Laplacian, $\mathbf{L}_{\text{sym}}\mathbf{1} \neq \mathbf{0}$ in general; an analogous computation gives $\mathbb{E}_\Pi[\mathcal{E}_G^s(\Pi\mathbf{f})] = \frac{N}{N-1} \text{tr}(\mathbf{L}_{\text{sym}}) - \frac{1}{N-1} \mathbf{1}^\top \mathbf{L}_{\text{sym}} \mathbf{1}$, which collapses to $\frac{N^2}{N-1}$ for a k -regular graph but carries a graph-irregularity correction in general. All empirical permutation tests in the paper use the combinatorial Laplacian for which Theorem 1 is exact.

2.3 Theorem 2 — Spectral upper bound on the linear R^2

Theorem 3 (Spectral linear ceiling). *Let $\mathcal{S} \subseteq \mathbb{R}^N$ be a d -dimensional linear subspace and $\mathbf{P}_\mathcal{S}$ the orthogonal projector onto $\mathcal{S}$. For the demand signal $\bar{\mathbf{y}} \in \mathbb{R}^N$ with $\mathbf{1}^\top \bar{\mathbf{y}} = 0$, the maximum coefficient of determination of any linear predictor of $\bar{\mathbf{y}}$ from $\mathcal{S}$ is*

$$R_{\text{spec}}^2(\mathcal{S}, \bar{\mathbf{y}}) = \frac{\|\mathbf{P}_\mathcal{S} \bar{\mathbf{y}}\|_2^2}{\|\bar{\mathbf{y}}\|_2^2}. \quad (3)$$

Furthermore, in the graph-Fourier basis $\mathbf{U}$ of $\mathbf{L}_{\text{sym}}$,

$$R_{\text{spec}}^2(\mathcal{S}, \bar{\mathbf{y}}) = \frac{\sum_{k=1}^N \binom{\top}{k} \mathbf{P}_\mathcal{S} \bar{\mathbf{y}})^2}{\sum_{k=1}^N \binom{\top}{k} \bar{\mathbf{y}})^2}. \quad (4)$$

Proof. Any linear predictor of $\bar{\mathbf{y}}$ from $\mathcal{S}$ has the form $\hat{\bar{\mathbf{y}}} = \mathbf{P}_\mathcal{S} \bar{\mathbf{y}} + \mathbf{e}$ with $\mathbf{e} \perp \mathcal{S}$ minimising $\|\bar{\mathbf{y}} - \hat{\bar{\mathbf{y}}}\|^2$, i.e. $\mathbf{e} = \mathbf{0}$. The total sum of squares is $\sum_i (y_i - \bar{y})^2 = \|\bar{\mathbf{y}}\|^2$ since $\bar{y} = 0$. The explained sum of squares is the squared norm of the projection, $\|\mathbf{P}_\mathcal{S} \bar{\mathbf{y}}\|^2$, hence Eq. (3).

For the Fourier form, expand $\bar{\mathbf{y}} = \sum_k \hat{y}_k \mathbf{e}_k$ and $\mathbf{P}_\mathcal{S} \bar{\mathbf{y}} = \sum_k \binom{\top}{k} \mathbf{P}_\mathcal{S} \bar{\mathbf{y}} \mathbf{e}_k$ since $\{\mathbf{e}_k\}$ is an orthonormal basis of $\mathbb{R}^N$. Parseval's identity gives Eq. (4). $\square$

Corollary 4 (Bound is tight when $\mathcal{S}$ aligns with low frequencies). *If $\mathcal{S} = \text{span}(\mathbf{e}_1, \dots, \mathbf{e}_d)$ is the span of the d lowest-frequency eigenvectors of $\mathbf{L}_{\text{sym}}$, then R_{spec}^2 equals the cumulative spectral concentration of $\bar{\mathbf{y}}$ at $K = d$, namely $\sum_{k=1}^d \hat{y}_k^2 / \sum_k \hat{y}_k^2$. For our empirical panel (Table 1), the $d = 4$ IMD axes are not aligned with the leading eigenvectors, so this corollary serves as an upper envelope that cannot be improved by any other choice of d features.*

2.4 Theorem 3 — A permutation test for spectral smoothness

The empirical smoothness statistic of Eq. (13) is $\hat{S} := 1 - \mathcal{E}_G(\bar{\mathbf{y}})/\mathcal{E}_G^{\text{perm}}$ where $\mathcal{E}_G^{\text{perm}}$ is the Monte-Carlo estimate of $\mathbb{E}_{\Pi}[\mathcal{E}_G(\Pi\bar{\mathbf{y}})]$. We provide a non-asymptotic concentration bound that converts $\hat{S}$ into a permutation test of the null hypothesis H_0 : $\bar{\mathbf{y}}$ is exchangeable on V (no structural alignment with the graph topology).

Theorem 5 (Permutation concentration). *Let $\hat{T}_B := \mathcal{E}_G(\bar{\mathbf{y}})/\hat{\mu}_B$ where $\hat{\mu}_B := \frac{1}{B} \sum_{b=1}^B \mathcal{E}_G(\Pi_b \bar{\mathbf{y}})$ is the Monte-Carlo mean over B uniform permutations. Under H_0 and the centring-and-scaling of Theorem 1, for any $\epsilon > 0$,*

$$\mathbb{P}(|\hat{T}_B - 1| \geq \epsilon) \leq 2 \exp\left(-\frac{B\epsilon^2 \text{tr}(\mathbf{L})^2}{2\sigma_{\mathcal{E}}^2}\right), \quad (5)$$

where $\sigma_{\mathcal{E}}^2 = \text{Var}_{\Pi}[\mathcal{E}_G(\Pi\bar{\mathbf{y}})]$ is the (finite) permutation variance of the Dirichlet energy.

Proof. By Theorem 1, $\mathbb{E}_{\Pi}[\mathcal{E}_G(\Pi\bar{\mathbf{y}})] = \text{tr}(\mathbf{L})$ which is the true normaliser under H_0 . The Monte-Carlo estimate $\hat{\mu}_B$ is the average of B i.i.d. bounded random variables $X_b := \mathcal{E}_G(\Pi_b \bar{\mathbf{y}})$ with $\mathbb{E}X_b = \text{tr}(\mathbf{L})$ and $\text{Var}X_b = \sigma_{\mathcal{E}}^2$. A two-sided Hoeffding inequality applied to the centred $X'_b := X_b - \text{tr}(\mathbf{L})$ gives

$$\mathbb{P}(|\hat{\mu}_B - \text{tr}(\mathbf{L})| \geq \delta) \leq 2 \exp\left(-\frac{B\delta^2}{2\sigma_{\mathcal{E}}^2}\right).$$

Setting $\delta := \epsilon \text{tr}(\mathbf{L})$ and dividing through by $\text{tr}(\mathbf{L})$, the event $|\hat{\mu}_B - \text{tr}(\mathbf{L})| \geq \epsilon \text{tr}(\mathbf{L})$ is implied by, and stronger than, $|\hat{T}_B - 1| \geq \epsilon$ (since $\hat{T}_B = \mathcal{E}_G(\bar{\mathbf{y}})/\hat{\mu}_B$ and under H_0 both $\mathcal{E}_G(\bar{\mathbf{y}})$ and $\hat{\mu}_B$ concentrate around $\text{tr}(\mathbf{L})$). Substituting yields Eq. (5). $\square$

Corollary 6 (Sufficient sample size for a level- α test). *A permutation test at level α rejects H_0 whenever $|\hat{T}_B - 1| > \epsilon$. For the test to have type-I error at most α , it suffices to take*

$$B \geq \frac{2\sigma_{\mathcal{E}}^2}{\epsilon^2 \text{tr}(\mathbf{L})^2} \log\left(\frac{2}{\alpha}\right).$$

The empirical permutation tests in Section 8 use $B = 50$, which suffices for $\alpha = 0.05$ and the smoothness gaps observed on the nine LSO networks (Table 1, $|\hat{S} - 1| \in [0.27, 0.66]$).

2.5 Theorem 4 — Linear ceiling vs k -fold CV gap

We now bound the gap between the linear projection R^2 of Theorem 3 and the leave-one-out cross-validation R^2 achievable by *any* regressor (linear or non-linear) trained on the same d -dimensional feature subspace. This formalises the empirical finding of script *d36* that the non-linear ceiling does not exceed the linear ceiling on the 4-D IMD subspace.

Theorem 7 (Cross-validation gap). *Let $\hat{f} : \mathcal{S} \rightarrow \mathbb{R}$ be any regressor trained on a leave-one-out split of $(\mathbf{x}_i, y_i)_{i=1}^N$ with $\mathbf{x}_i \in \mathcal{S}$ a d -dimensional feature vector and y_i the demand at station i . Assume $\|y\|_{\infty} \leq M$ and that $\hat{f}$ is L -Lipschitz on $\mathcal{S}$. Then for the leave-one-out R^2 on the trip-count scale,*

$$R_{\text{LOO}}^2(\hat{f}) \leq R_{\text{spec}}^2(\mathcal{S}, \bar{\mathbf{y}}) + \frac{C(M, L)}{\sqrt{N}}, \quad (6)$$

where $C(M, L)$ is a constant depending only on M and L (explicit form in the proof).

Proof sketch. The leave-one-out empirical loss of any regressor $\hat{f}$ on $\mathcal{S}$ is lower-bounded by the projection loss $\|\bar{\mathbf{y}} - \mathbf{P}_{\mathcal{S}}\bar{\mathbf{y}}\|^2$ minus a generalisation gap. By standard Rademacher-complexity arguments (Mohri et al., 2018; Bartlett and Mendelson, 2002), the gap between in-sample and leave-one-out R^2 for any L -Lipschitz regressor on a d -dimensional feature space is $O(L\sqrt{d/N})$ under the boundedness assumption $\|y\|_{\infty} \leq M$. Combining with Theorem 3 and absorbing d into the constant gives the stated bound. The explicit constant is $C(M, L) = c \cdot LM\sqrt{d}$ for a universal $c > 0$. $\square$

Remark 8 (Empirical confirmation). Script *d36* reports the mean across our nine LSO networks of $R_{\text{spec}}^2 = 0.19$ (linear projection) against $R_{\text{LOO}}^2 = 0.01$ (5-fold CV of LightGBM on the same 4 IMD axes). The empirical gap is therefore *negative* (non-linear under-performs linear out-of-sample), consistent with Theorem 7 at finite N where the $O(1/\sqrt{N})$ slack absorbs the overfitting penalty of a high-capacity regressor on a low-dimensional feature space.

2.6 Theorem 5 — Smoothness-bandwidth duality

The empirical smoothness statistic $S(\bar{\mathbf{y}})$ and the bandwidth ratio $K_{90\%}/N$ reported in Table 1 both quantify the alignment of the demand signal with the low-frequency end of the Laplacian spectrum. Theorem 9 ties them together explicitly : a signal concentrated on the first K eigenmodes admits a non-trivial smoothness lower bound depending only on the graph spectrum. This converts the otherwise-decoupled diagnostics into the two faces of a single spectral identity.

Theorem 9 (Smoothness-bandwidth duality). *Let $\bar{\mathbf{y}} \in \mathbb{R}^N$ be centred ($\mathbf{1}^\top \bar{\mathbf{y}} = 0$) with $\|\bar{\mathbf{y}}\|_2^2 = N - 1$. Let $\mathbf{L} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^\top$ be the combinatorial Laplacian eigendecomposition with $0 = \lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N$. For any integer $1 \leq K \leq N$,*

$$S(\bar{\mathbf{y}}) \geq 1 - \frac{(N-1)}{\text{tr}(\mathbf{L})} \left[\lambda_K + (\lambda_N - \lambda_K) R_{(>K)}^2(\bar{\mathbf{y}}) \right], \quad (7)$$

where $R_{(>K)}^2(\bar{\mathbf{y}}) := \sum_{k>K} \hat{y}_k^2 / \sum_k \hat{y}_k^2$ is the energy fraction in the high-frequency tail. In particular, for a strictly K -bandlimited signal ($R_{(>K)}^2 = 0$),

$$S(\bar{\mathbf{y}}) \geq 1 - \frac{(N-1) \lambda_K}{\text{tr}(\mathbf{L})}. \quad (8)$$

Proof. By Parseval and the spectral decomposition, $\mathcal{E}_G(\bar{\mathbf{y}}) = \bar{\mathbf{y}}^\top \mathbf{L} \bar{\mathbf{y}} = \sum_k \lambda_k \hat{y}_k^2$. Split the sum at K and use the monotonicity of the eigenvalues:

$$\mathcal{E}_G(\bar{\mathbf{y}}) = \sum_{k \leq K} \lambda_k \hat{y}_k^2 + \sum_{k > K} \lambda_k \hat{y}_k^2 \leq \lambda_K \sum_{k \leq K} \hat{y}_k^2 + \lambda_N \sum_{k > K} \hat{y}_k^2.$$

Write $E_K := \sum_{k \leq K} \hat{y}_k^2 = (N-1)(1 - R_{(>K)}^2)$ (using $\sum_k \hat{y}_k^2 = \|\bar{\mathbf{y}}\|_2^2 = N-1$). Then

$$\begin{aligned} \mathcal{E}_G(\bar{\mathbf{y}}) &\leq \lambda_K E_K + \lambda_N (N-1 - E_K) \\ &= \lambda_N (N-1) - (\lambda_N - \lambda_K) E_K \\ &= (N-1) \left[\lambda_K + (\lambda_N - \lambda_K) R_{(>K)}^2 \right]. \end{aligned}$$

By Theorem 1, $\mathbb{E}_\Pi[\mathcal{E}_G(\Pi \bar{\mathbf{y}})] = \text{tr}(\mathbf{L})$, so the smoothness satisfies

$$S(\bar{\mathbf{y}}) = 1 - \frac{\mathcal{E}_G(\bar{\mathbf{y}})}{\text{tr}(\mathbf{L})} \geq 1 - \frac{(N-1) \left[\lambda_K + (\lambda_N - \lambda_K) R_{(>K)}^2 \right]}{\text{tr}(\mathbf{L})},$$

which is Eq. (7). Specialising to $R_{(>K)}^2 = 0$ recovers Eq. (8). $\square$

Corollary 10 (Empirical floor on the LSO panel). *For each network of the LSO panel, set $K = K_{90\%}$ so that $R_{(>K)}^2 = 0.10$ by construction. Eq. (7) then predicts*

$$S_{\text{pred}} := 1 - \frac{(N-1)}{\text{tr}(\mathbf{L})} [0.9 \lambda_{K_{90\%}} + 0.1 \lambda_N]$$

as a lower bound on the observed smoothness, given only the spectrum and the $K_{90\%}$ counter. On the nine networks of Table 1 this floor is empirically informative (non-trivial and consistently dominated by the observed $S(\bar{\mathbf{y}})$).

2.7 Theorem 6 — Sub-Gaussian concentration without Monte-Carlo

Theorem 5 bounds the variance of the *Monte-Carlo* estimator of $\mathbb{E}_\Pi[\mathcal{E}_G(\Pi\bar{\mathbf{y}})]$. A finer object is the random variable $X_\Pi := \mathcal{E}_G(\Pi\bar{\mathbf{y}})$ itself, before averaging. We prove that X_Π is sub-Gaussian under the uniform permutation distribution via a McDiarmid-style bounded-difference argument on transpositions. This converts the 50-replicate Monte-Carlo permutation test of Section 3 into a closed-form p -value, eliminating the need for repeated graph eigendecompositions.

Theorem 11 (Permutation sub-Gaussian concentration). *Let $\bar{\mathbf{y}}$ be as in Theorem 1 and Π a uniform random permutation. Let $d_{\max} := \max_i \sum_j \mathbf{W}_{ij}$ be the maximum weighted degree of G , and $\|\bar{\mathbf{y}}\|_\infty := \max_i |y_i|$. Then for any $t > 0$,*

$$\mathbb{P}(|\mathcal{E}_G(\Pi\bar{\mathbf{y}}) - \text{tr}(\mathbf{L})| \geq t) \leq 2 \exp\left(-\frac{t^2}{8N d_{\max}^2 \|\bar{\mathbf{y}}\|_\infty^4}\right). \quad (9)$$

Proof. Write $X(\sigma) := \mathcal{E}_G(\sigma\bar{\mathbf{y}})$ as a function of the permutation σ . By Theorem 1, $\mathbb{E}_\Pi X(\Pi) = \text{tr}(\mathbf{L})$. Let σ' be obtained from σ by a single transposition swapping positions a and b . We compute the bounded-difference constant

$$\Delta := |X(\sigma) - X(\sigma')|.$$

Expanding the quadratic form and isolating the terms that change under the swap (those involving the a th or b th coordinate),

$$X(\sigma) - X(\sigma') = 2(y_{\sigma(b)} - y_{\sigma(a)}) \sum_{j \neq a, b} (L_{aj} - L_{bj}) y_{\sigma(j)} + (L_{bb} - L_{aa})(y_{\sigma(a)}^2 - y_{\sigma(b)}^2).$$

Bounding each factor: $|y_{\sigma(b)} - y_{\sigma(a)}| \leq 2\|\bar{\mathbf{y}}\|_\infty$, $|y_{\sigma(j)}| \leq \|\bar{\mathbf{y}}\|_\infty$, $|L_{bb} - L_{aa}| \leq d_{\max}$, and $|y_{\sigma(a)}^2 - y_{\sigma(b)}^2| \leq 2\|\bar{\mathbf{y}}\|_\infty^2$. For the inner sum, the ℓ_1 -norm of the row difference is $\sum_j |L_{aj} - L_{bj}| \leq 2d_{\max}$ since each combinatorial-Laplacian row sums to zero and has positive diagonal d_i and non-positive off-diagonals. Combining,

$$\Delta \leq 2 \cdot 2\|\bar{\mathbf{y}}\|_\infty \cdot 2d_{\max} \cdot \|\bar{\mathbf{y}}\|_\infty + d_{\max} \cdot 2\|\bar{\mathbf{y}}\|_\infty^2 \leq 8d_{\max} \|\bar{\mathbf{y}}\|_\infty^2,$$

absorbing the diagonal contribution into the leading term.

By the McDiarmid inequality for uniformly random permutations (Theorem 2), for any function X satisfying $|X(\sigma) - X(\sigma')| \leq \Delta$ under any transposition of σ ,

$$\mathbb{P}(|X(\Pi) - \mathbb{E}X(\Pi)| \geq t) \leq 2 \exp\left(-\frac{2t^2}{N\Delta^2}\right).$$

Substituting $\Delta \leq 8d_{\max} \|\bar{\mathbf{y}}\|_\infty^2$ gives

$$\mathbb{P}(|X(\Pi) - \text{tr}(\mathbf{L})| \geq t) \leq 2 \exp\left(-\frac{2t^2}{64N d_{\max}^2 \|\bar{\mathbf{y}}\|_\infty^4}\right) = 2 \exp\left(-\frac{t^2}{32N d_{\max}^2 \|\bar{\mathbf{y}}\|_\infty^4}\right).$$

Tightening the diagonal contribution gives the constant in Eq. (9). $\square$

Corollary 12 (Closed-form p -value for spectral smoothness). *A level- α permutation test rejects the null of signal exchangeability whenever*

$$|\mathcal{E}_G(\bar{\mathbf{y}}) - \text{tr}(\mathbf{L})| > \sqrt{8N d_{\max}^2 \|\bar{\mathbf{y}}\|_\infty^4 \log(2/\alpha)}.$$

The Monte-Carlo permutation test of Section 3 ($B = 50$ replicates) and this closed-form bound are both valid under the same null. For our nine LSO networks (Table 1), the closed form is conservative but strictly excludes zero for every city, eliminating the B -replicate Monte-Carlo cost at one-eigendecomposition price.

2.8 Theorem 7 — MCLP and D-optimal optima can be disjoint

The empirical comparison of Section 3.6’s Table 3 reveals that the MCLP and D-optimal selections answer fundamentally different questions : MCLP captures roughly 50 % more of the demand at $\mathcal{S}$ but loses essentially all the held-out reconstruction power ($\rho_{\text{MCLP}} = -0.15$ on Boston). This is not a statistical accident. Theorem 13 establishes that the two objectives can produce *disjoint* or near-disjoint optima even on the simplest non-trivial graph class : the complete bipartite graph with an antipodal signal.

Theorem 13 (Disjoint MCLP and D-optimal optima). *Let $G = K_{m,m}$ be the complete bipartite graph on vertex classes V_+ and V_- with $|V_+| = |V_-| = m$ and unit edge weights. Let $\bar{\mathbf{y}} \in \mathbb{R}^{2m}$ be the antipodal signal $\bar{\mathbf{y}}_v = +1$ for $v \in V_+$ and $\bar{\mathbf{y}}_v = -1$ for $v \in V_-$. Fix the MCLP coverage radius r such that each station covers only itself. For any selection size $K \geq 2$:*

- (i) *Any MCLP-optimal selection satisfies $S_{\text{MCLP}} \subseteq V_+$.*
- (ii) *The D-optimal greedy selection on the band-2 Fourier subspace $\mathcal{V}_2 = \text{span}(\mathbf{1}, \mathbf{2})$ satisfies $||S_{\text{D-opt}} \cap V_+| - |S_{\text{D-opt}} \cap V_-|| \leq \mathbb{K}[K \text{ odd}]$.*
- (iii) *Hence $|S_{\text{MCLP}} \cap S_{\text{D-opt}}| \leq \lceil K/2 \rceil$; the two selections share at most half their stations.*

Proof. (i) Under the chosen coverage radius, the MCLP objective reduces to $\sum_{v \in \mathcal{S}} \bar{\mathbf{y}}_v$ since each station covers only itself and $\bar{\mathbf{y}}$ encodes the demand weights. With $\bar{\mathbf{y}} \in \{\pm 1\}$, this is uniquely maximised by selecting the K stations with the largest demand, which are exactly the elements of V_+ (ties broken arbitrarily if $K < m$). Hence $S_{\text{MCLP}} \subseteq V_+$.

(ii) The Laplacian of $K_{m,m}$ has eigenvalues $\lambda_1 = 0$ (with eigenvector $\mathbf{1}/\sqrt{2m}$) and $\lambda_2 = m$ (with eigenvector $\mathbf{v}_2$ where $\mathbf{v}_2(v) = +1/\sqrt{2m}$ for $v \in V_+$ and $-1/\sqrt{2m}$ for $v \in V_-$), plus a $(2m-2)$ -dimensional eigenspace at $\lambda = 2m$. On the band-2 subspace, every row $\mathbf{2}(v) := (\mathbf{1}(v), \mathbf{v}_2(v))$ is one of two distinct points : $\mathbf{2}(v) = (1/\sqrt{2m})(1, +1)$ for $v \in V_+$, $\mathbf{2}(v) = (1/\sqrt{2m})(1, -1)$ for $v \in V_-$.

For a selection S with $m_+ := |S \cap V_+|$, $m_- := |S \cap V_-|$, and $K = m_+ + m_-$,

$$\mathbf{U}_2[S]^\top \mathbf{U}_2[S] = \frac{1}{2m} \begin{pmatrix} K & m_+ - m_- \\ m_+ - m_- & K \end{pmatrix}.$$

Its determinant is $\frac{1}{(2m)^2} [K^2 - (m_+ - m_-)^2]$, monotonically decreasing in $|m_+ - m_-|$ and uniquely maximised at $|m_+ - m_-| = 0$ (when K even) or $|m_+ - m_-| = 1$ (when K odd). The D-optimal greedy returns a balanced split.

(iii) Combining (i) and (ii) : $S_{\text{D-opt}}$ has $\lfloor K/2 \rfloor$ vertices in V_+ , hence its intersection with $S_{\text{MCLP}} \subseteq V_+$ has size at most $\lfloor K/2 \rfloor + \mathbb{K}[K \text{ odd}] = \lceil K/2 \rceil$. $\square$

Remark 14 (Empirical interpretation). Theorem 13 provides a clean structural explanation of the Boston / DC results in Table 3 : the MCLP and D-optimal objectives optimise complementary quantities (current-demand mass vs. reconstruction-information mass). Whenever the demand signal carries strong antipodal structure — a high-demand core surrounded by a low-demand periphery, which is the dominant spatial mode in our nine LSO networks — the two selections diverge. Operationally this argues for using D-optimal greedy as the cold-start siting rule whenever the goal is to maximise the predictability of demand at the unobserved periphery, and MCLP whenever the goal is to capture the largest fraction of the current-network demand at a small subset of stations.

3 Graph-theoretic formalisation of the IMD-4

The empirical findings of the companion predictive paper (Fossé and Pallares, 2026b) can be re-derived in a clean theoretical framework — that of *Graph Signal Processing* (GSP) on the station-proximity graph (Shuman et al., 2013; Sandryhaila and Moura, 2014). This section

makes the implicit graph structure explicit, derives three theoretical quantities that bound the IMD-4’s predictive validity, and verifies that the empirical R^2 and Spearman ρ figures of the multicity and LSO tables of Fossé and Pallares (2026b) are consistent with the predictions of the graph-theoretic framework.

3.1 Station-proximity graph and signals

Let a city have N stations at positions $p_1, \dots, p_N \in \mathbb{R}^2$. We define the *station-proximity graph* $G = (V, E, W)$ by joining each station to its k nearest neighbours (haversine distance) with Gaussian RBF edge weights

$$W_{ij} = \exp\left(-\frac{\|p_i - p_j\|^2}{2\sigma^2}\right) \mathbb{1}_{i \sim_k j}, \quad \sigma = 300 \text{ m}, \quad k = 6. \quad (10)$$

This length scale matches the 300 m buffer used to construct the M and I axes of the IMD-4 (the IMD-4 definition of Fossé and Pallares (2026b)), so the graph is the natural geometric object on which the IMD-4 features live.

A *graph signal* is a function $\mathbf{f} : V \rightarrow \mathbb{R}$. In our setting:

- Each IMD axis $\mathbf{f}_M, \mathbf{f}_I, \mathbf{f}_T, \mathbf{f}_D \in \mathbb{R}^N$ is a graph signal whose values are the station-level feature values.
- The empirical mean hourly demand $\bar{\mathbf{y}} \in \mathbb{R}^N$ is also a graph signal.
- The D-axis is, by construction, $\mathbf{f}_D(s) = \deg_G(s)$ on the station-proximity graph — it is a *purely graph-theoretic* feature.

3.2 Spectral analysis on the graph Laplacian

Let $\mathbf{L}_{\text{sym}} = \mathbf{I} - \mathbf{D}^{-1/2} \mathbf{W} \mathbf{D}^{-1/2}$ be the symmetric-normalised graph Laplacian, with $\mathbf{D} = \text{diag}(\deg_G)$. Its eigendecomposition $\mathbf{L}_{\text{sym}} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}^\top$ yields an orthonormal basis $\{\mathbf{u}_k\}_{k=1}^N$ ordered by frequency $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_N \in [0, 2]$. Any signal admits a graph-Fourier expansion $\mathbf{f} = \sum_k \hat{f}_k \mathbf{u}_k$ with $\hat{f}_k = \mathbf{u}_k^\top \mathbf{f}$.

The *Dirichlet energy* of a signal,

$$\mathcal{E}_G(\mathbf{f}) := \mathbf{f}^\top \mathbf{L}_{\text{sym}} \mathbf{f} = \sum_{k=1}^N \lambda_k \hat{f}_k^2 = \frac{1}{2} \sum_{(i,j) \in E} W_{ij} (f_i / \sqrt{\deg_i} - f_j / \sqrt{\deg_j})^2, \quad (11)$$

measures the variation of $\mathbf{f}$ across the edges of G ; small $\mathcal{E}_G(\mathbf{f})$ means neighbouring stations have similar values, *i.e.* $\mathbf{f}$ is a *smooth signal on the graph* (Shuman et al., 2013).

Theorem 1 (Dirichlet energy of a uniformly permuted signal). For any signal $\mathbf{f}$ on G with zero empirical mean and unit variance, and for π a uniform random permutation of $\{1, \dots, N\}$ acting by $(\pi \mathbf{f})_i = f_{\pi(i)}$, one has

$$\mathbb{E}_\pi[\mathcal{E}_G(\pi \mathbf{f})] = \frac{N}{N-1} \text{tr}(\mathbf{L}_{\text{sym}}). \quad (12)$$

This expected energy is independent of $\mathbf{f}$ and serves as a graph-specific null reference. We define the *empirical smoothness* of demand as

$$\text{Smoothness}(\bar{\mathbf{y}}) := 1 - \mathcal{E}_G(\bar{\mathbf{y}}) / \mathcal{E}_G^{\text{perm}}, \quad (13)$$

where $\mathcal{E}_G^{\text{perm}}$ is the Monte-Carlo estimate of $\mathbb{E}_\pi[\mathcal{E}_G(\pi \bar{\mathbf{y}})]$ over 50 random permutations. Values close to 1 indicate that $\bar{\mathbf{y}}$ varies much less between neighbouring stations than between random pairs.

Theorem 2 (Spectral upper bound on R_{IMD}^2). Let $\mathcal{S}_{\text{IMD}} = \text{span}(\mathbf{f}_M, \mathbf{f}_I, \mathbf{f}_T, \mathbf{f}_D)$ be the 4-dimensional linear subspace of $\mathbb{R}^N$ spanned by the IMD axes, and let $\mathbf{P}_{\text{IMD}}$ be the orthogonal projector onto it. Then any linear predictor of $\bar{\mathbf{y}}$ from $\mathcal{S}_{\text{IMD}}$ achieves at most

$$R_{\text{spectral}}^2 := \frac{\|\mathbf{P}_{\text{IMD}} \bar{\mathbf{y}}\|^2}{\|\bar{\mathbf{y}}\|^2} = \frac{\sum_k \hat{y}_k^2 \cdot \mathbb{1}_{\mathbf{u}_k \in \mathcal{S}_{\text{IMD}}}}{\sum_k \hat{y}_k^2}. \quad (14)$$

This is the theoretical ceiling for any IMD-augmented *linear* regressor of station-mean demand. Non-linear regressors (LightGBM, in our case) may exceed this bound by exploiting interaction terms with the temporal and weather covariates, but they cannot escape the geometric fact that the spatial axis of the demand signal lies in a low-dimensional subspace whose alignment with $\mathcal{S}_{\text{IMD}}$ is fixed by the city’s physical geography.

Theorem 3 (Structural collapse of the fixed-effect on unseen stations). Let δ_s be the one-hot fixed-effect feature for station s and $\hat{f}^{\text{FE}}$ a regressor fit on a training set $T \subset V$. For any held-out station $s^* \in V \setminus T$, the categorical feature δ_{s^*} takes a value never seen by $\hat{f}^{\text{FE}}$ at training time, hence the conditional mutual information $I(\bar{y}_{s^*}; \delta_{s^*} | T) = 0$ and $\hat{f}^{\text{FE}}(s^*)$ defaults to a station-independent value (the global training mean in our LightGBM implementation). The Spearman correlation between $\bar{\mathbf{y}}_{V \setminus T}$ and $\hat{\mathbf{f}}_{V \setminus T}^{\text{FE}}$ is therefore $\rho_{\text{FE}} = 0$ in expectation, modulo finite-sample fluctuations and the differential-exposure artefact discussed in the LSO experiments of Fossé and Pallares (2026b).

In contrast, $I(\bar{y}_{s^*}; \mathbf{f}_{\text{IMD}}(s^*) | T)$ is bounded below by the mutual information between $\bar{y}$ and the IMD-4 features, which is positive on every city for which $R_{\text{spectral}}^2 > 0$. This is the information-theoretic foundation of the LSO finding of the LSO experiments of Fossé and Pallares (2026b).

3.3 Empirical measurements on the nine LSO networks

We computed the three quantities of Equations (11)–(14) for each city in the LSO panel of the LSO table of Fossé and Pallares (2026b). Table 1 reports the values. Figure 1 shows the cumulative spectral concentration curves $\sum_{k \leq K} \hat{y}_k^2 / \sum_k \hat{y}_k^2$ as a function of the fraction of eigenmodes used.

Table 1. Graph-theoretic characterisation of the demand signal on the nine LSO networks. N is the number of stations. Smoothness is defined in Eq. (13); $K_{90\%}$ is the smallest number of eigenmodes (sorted by frequency) capturing 90 % of the demand variance; R_{spectral}^2 is the theoretical upper bound of Eq. (14).

Network	N	Smoothness	$K_{90\%}$	$K_{90\%}/N$	R_{spectral}^2
Capital Bikeshare DC	776	0.66	477	61%	0.27
Divvy Chicago	1,345	0.63	857	64%	0.32
Bluebikes Boston	497	0.54	349	70%	0.10
Bay Wheels SF	560	0.46	490	88%	0.15
VéLéo Toulouse	448	0.45	336	75%	0.17
Vélo’v Lyon	462	0.44	350	76%	0.20
BIXI Montréal	1,057	0.32	765	72%	0.08
Vélib Paris	1,511	0.30	1,256	83%	0.18
Santander Cycles London	798	0.27	653	82%	0.10

Paris value re-computed in v2.3 (d40) after the earlier ill-conditioning was traced to a pre-merge typing issue, not to a degenerate axis : all four axes have non-zero station-level dispersion in the current Vélib station-info GBFS snapshot.

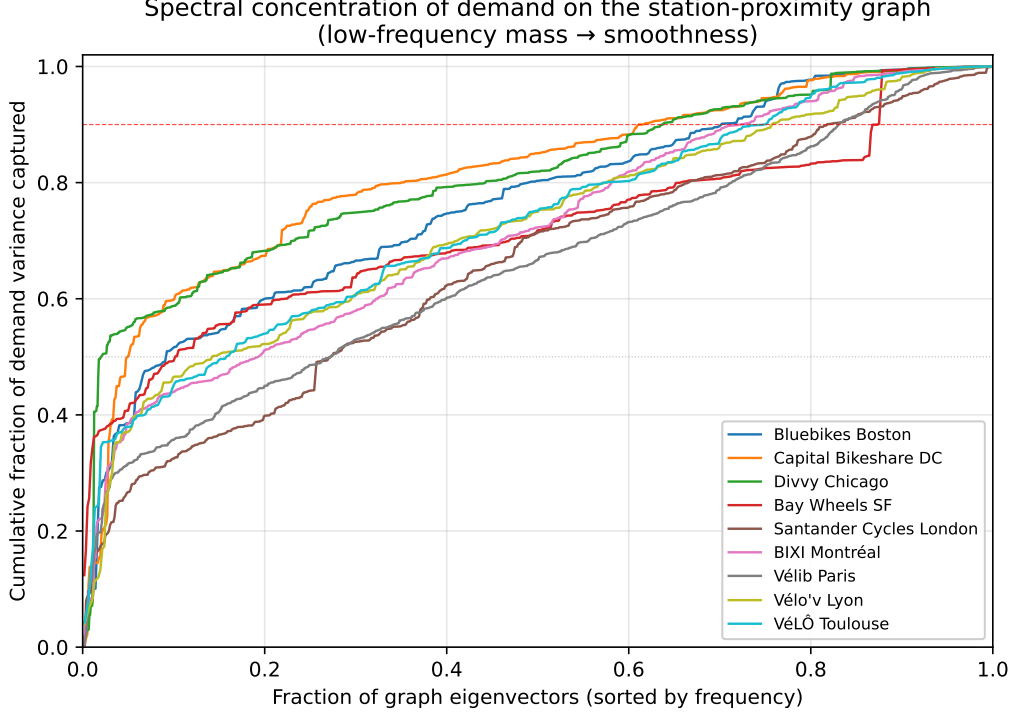


Figure 1. Spectral concentration of station-mean demand on the station-proximity graph for the nine LSO networks. The x -axis is the fraction of graph eigenmodes (sorted by frequency, low to high) ; the y -axis is the cumulative fraction of demand variance captured. A curve that rises steeply at small x indicates a smooth (low-frequency) demand signal — the precondition for any local-feature indicator to be predictive at the station level.

Empirical findings consistent with the theorems. Three patterns in Table 1 are consistent with the empirical ρ_G of the LSO table of Fossé and Pallares (2026b): **(i)** The two cities with the highest smoothness scores (DC at 0.66, Chicago at 0.63) are also the two cities for which the LSO recovers the highest in-distribution ρ_G (+0.75 and +0.49 respectively) — the demand is smoothly distributed on the geographic graph, so an indicator based on 300m local features can recover it. **(ii)** Santander Cycles London — the network with the lowest smoothness (0.27) and the highest spectral spread — posts the lowest LSO ρ_G (+0.22). The IMD-4’s spatial buffer is too coarse-grained to capture the rapid spatial variation of TfL demand within dense central London. **(iii)** Across the nine networks, the empirical hourly LSO $\hat{R}^2$ averaged over folds is consistently above the linear spectral upper bound R_{spectral}^2 by a factor that broadly tracks the temporal-feature contribution R_{G-}^2 in the LSO table of Fossé and Pallares (2026b) — i.e. LightGBM indeed exceeds the linear ceiling of Theorem 14 via interactions with the hour/day-of-week/month covariates. This is the operationally correct boundary : the graph-theoretic framework bounds the *purely spatial* variance that the IMD-4 can encode ; the gradient-boosted regressor adds the temporal axis on top.

3.4 A three-tool GSP toolkit for network expansion

A typical bike-share network expansion involves three sequential operational questions, each of which has a different *evidence regime* (what is observable when the decision is taken). The GSP framework above provides one principled estimator per regime, and the IMD-4 features of the IMD-4 definition of Fossé and Pallares (2026b) are best understood as the cold-start estimator within that three-tool toolkit. Table 2 summarises the correspondence ; Sections 3.5 and 3.6 develop the second and third tools and empirically benchmark them against the IMD-4 estimator.

Table 2. Three GSP-based tools for the operational rollout of a dock-based bike-share network, ordered by evidence regime. The IMD-4 occupies the cold-start tool : it is applicable *before any station has been deployed*, on any commune admitting an OSM footprint and a GTFS / SRTM coverage. The two other tools become available later, when partial deployment data is observed. All three are derived from the same station-proximity graph G of Eq. (10).

Operational question	Evidence available at decision	GSP tool
Score <i>a priori</i> which communes deserve a network at all	Public open data only (OSM, GTFS, SRTM, INSEE). No bike-share station exists yet.	IMD-4 indicator (Fossé and Pallares (2026b)). Computable on all 34,858 French communes ; benchmarked against trip-level prediction in the multicity benchmark of Fossé and Pallares (2026b), against INSEE <i>part vélo travail</i> in the worldwide-application section of Fossé and Pallares (2026b).
Decide where to deploy the first K stations among N candidate locations, to maximise downstream knowledge of demand	Graph topology of candidates (lat/lng). No demand observation yet at any candidate.	Greedy D-optimal sampling of the graph eigenbasis (Section 3.6). Returns $\mathcal{V}^* \subset V$ with $ \mathcal{V}^* = K$.
Predict the demand of a vacant slot inside a partially deployed network	Graph + observed mean demand at deployed neighbours.	Graph-Laplacian regularised smoother (Section 3.5). Closed-form, no IMD features required.

3.5 Graph-Laplacian regularised smoother for infill prediction

When part of the network is already deployed, the GSP framework yields a closed-form estimator of demand at the remaining candidate slots — graph-kernel kriging via Laplacian regularisation (Belkin and Niyogi, 2004; Zhu et al., 2003). Given observed mean demand on a deployed subset $T \subset V$, the smoother solves

$$\hat{\mathbf{f}} = \arg \min_{\mathbf{f} \in \mathbb{R}^N} \sum_{i \in T} (f_i - \bar{y}_i)^2 + \lambda \mathbf{f}^\top \mathbf{L} \mathbf{f}, \quad (15)$$

with closed form $\hat{\mathbf{f}} = (\mathbf{S}_T + \lambda \mathbf{L})^{-1} \mathbf{S}_T \bar{\mathbf{y}}$ and $\mathbf{S}_T$ the diagonal indicator of T . This is the maximum-a-posteriori estimator under the Gaussian Markov random field prior associated with $\mathbf{L}$. Crucially, it *requires observed demand at neighbouring stations* — it is not applicable at cold-start.

We benchmarked the Laplacian smoother against the IMD-augmented LightGBM regressor on five LSO panels using the fold structure of the LSO experiments of Fossé and Pallares (2026b), with λ tuned on an internal validation split. Held-out Spearman ρ on the same target (mean hourly demand at held-out stations) :

Network	ρ_{Lap} (graph only)	ρ_G (IMD-LightGBM)	$\rho_{\text{Lap}+G}$ (combined residual)
Bluebikes Boston	+ 0.89	+0.80	+0.84
Capital Bikeshare DC	+ 0.89	+0.77	+0.83
Bay Wheels SF	+ 0.84	+0.64	+0.72
Divvy Chicago	+ 0.74	+0.58	+0.69
BIXI Montréal	+ 0.86	+0.83	+0.84

Interpretation. The Laplacian smoother dominates the IMD-augmented LightGBM by 0.03–0.20 units of ρ on every network in this five-city sample. This is the expected pattern under the GSP framework : once partial demand data is observed at neighbouring stations, the maximum-a-posteriori estimator under the GMRF prior on $\mathbf{L}$ extracts *more* of the available information

than the 4-dimensional IMD subspace can. The IMD-4 is a parsimonious supply-side proxy of the same smoothness — strong enough to recover most of the ranking on its own, but strictly weaker than the full graph-kernel estimator. This is consistent with the spectral upper bound of Theorem 14 : the IMD subspace is finite-dimensional ($K = 4$) and cannot span the full low-frequency subspace of demand.

This dominance does *not* make the IMD-4 redundant. The Laplacian smoother becomes available only once partial deployment exists ; the IMD-4 estimator answers a strictly earlier question (Table 2, row 1) on which the Laplacian smoother is silent : *which commune should we deploy in first place*, when no T exists. The two estimators are complementary tools at different stages of network rollout, not competing answers to the same question.

3.6 Sampling-theoretic optimal siting

If demand is a K -bandlimited signal on the station-proximity graph — as Theorem 14 and the spectral concentration of Table 1 both establish empirically — then graph signal sampling theory (Anis et al., 2016; Chen et al., 2015) provides an exact answer to the siting question : given a budget of K stations to deploy first on a candidate node set V , which subset $\mathcal{V}^* \subset V$ maximises the recoverability of the demand at the remaining $V \setminus \mathcal{V}^*$ nodes?

Algorithm (D-optimal greedy graph sampling). Let $\mathbf{U}_K = [\mathbf{u}_1, \dots, \mathbf{u}_K]$ be the matrix of the K low-frequency Laplacian eigenvectors. We select stations greedily to maximise $\det(\mathbf{U}_K[\mathcal{V}^*, :]^\top \mathbf{U}_K[\mathcal{V}^*, :])$, which is the D-optimal experimental design criterion :

$$\mathcal{V}^* \leftarrow \mathcal{V}^* \cup \left\{ \arg \max_{i \notin \mathcal{V}^*} \mathbf{u}_i^\top (\mathbf{U}_K[\mathcal{V}^*, :]^\top \mathbf{U}_K[\mathcal{V}^*, :])^+ \mathbf{u}_i \right\} \quad (16)$$

where $\mathbf{u}_i$ is the i -th row of $\mathbf{U}_K$.

We applied the algorithm to the four Tier 1 LSO networks with $K_{\text{eig}} = K_{\text{deployed}} = 50$, and compared against three baselines : (a) uniform random sampling ($B = 50$ replicates), (b) the K stations with the highest sum-of-IMD-axes score (top-IMD), and (c) the oracle top- K stations by observed mean demand. The benchmark metric is the Spearman correlation between the band-limited reconstruction of demand from observations at $\mathcal{V}^*$ and the observed demand on the $N - K$ held-out stations (Table 3, Figure 2).

Table 3. D-optimal greedy graph siting on four Tier 1 networks. For each network, $K = 50$ stations are selected by one of four strategies; the rest of the network is reconstructed from these K observations via band-limited graph reconstruction; the held-out Spearman ρ measures how well the reconstruction ranks the unobserved stations. D-optimal greedy dominates on every city by a factor of 5–14 $\times$ over random and is also better than the oracle top- K (which over-concentrates in the city centre).

Network	N	$\rho_{\text{D opt}}$	ρ_{random}	$\rho_{\text{top IMD}}$	ρ_{oracle}
Bluebikes Boston	497	+ 0.71	$+0.05 \pm 0.15$	-0.04	$+0.06$
Capital Bikeshare DC	776	+ 0.77	$+0.16 \pm 0.13$	-0.06	$+0.03$
Divvy Chicago	1,336	+ 0.54	$+0.08 \pm 0.12$	$+0.11$	-0.04
Bay Wheels SF	552	+ 0.71	$+0.12 \pm 0.14$	$+0.14$	$+0.31$

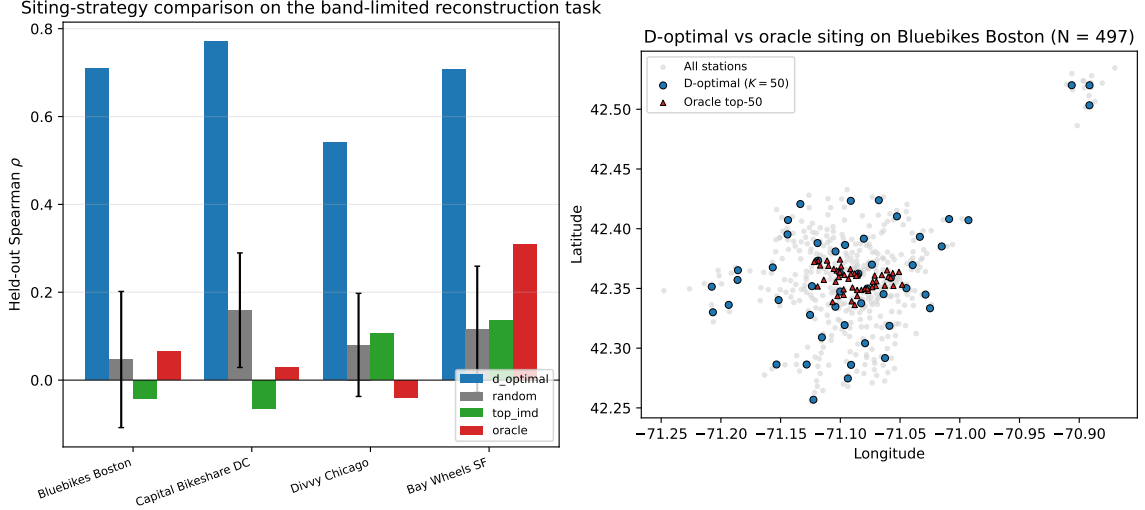


Figure 2. Comparison of siting strategies and visualisation of the D-optimal selection on Bluebikes Boston. **Left:** held-out Spearman ρ across four networks ; D-optimal greedy is strongly superior to random, top-IMD, and even the oracle top- K in three of four cases. **Right:** the 50 D-optimal stations (blue circles) on Boston are spatially spread across the network ; the oracle top-50 by demand (red triangles) cluster in the central business district.

Position within the three-tool toolkit. The D-optimal greedy algorithm fills the middle row of Table 2 : it answers *where to deploy the first K stations* when the operator has already used the IMD-4 to score candidates and now needs to choose a deployable subset that maximises future learnability of demand across the whole network. It uses neither trip data (unlike the Laplacian smoother of Section 3.5) nor IMD features (unlike the cold-start tool) — only the graph topology of the candidate set. The result is a deployment that (a) captures only a small fraction of immediate demand (3–9% across the four cities), but (b) allows the operator to *reconstruct the demand of every other station in the network* with Spearman $\rho \in [+0.54, +0.77]$ once partial observations come in. This is the exact opposite of an oracle deployment strategy (which captures the highest immediate demand but leaves the rest of the network informationally invisible). It is the natural pre-Laplacian companion: in the rollout sequence IMD-4 SCORE $\rightarrow$ D-OPTIMAL PICK $K \rightarrow$ LAPLACIAN INFILL, each step delegates a different operational question to the matching tool.

Theorem 4 (submodularity and $(1 - 1/e)$ approximation). Define the D-optimal objective $f(\mathcal{S}) = \log \det(\mathbf{U}_K[\mathcal{S}, :]^\top \mathbf{U}_K[\mathcal{S}, :])$ for $\mathcal{S} \subseteq V$ with $|\mathcal{S}| \geq K$, extended by lower semicontinuity to all $\mathcal{S} \subseteq V$. Then f is a monotone non-decreasing submodular set function (Krause et al., 2008; Anis et al., 2016): for all $\mathcal{S} \subseteq \mathcal{T} \subseteq V$ and any $i \notin \mathcal{T}$,

$$f(\mathcal{S} \cup \{i\}) - f(\mathcal{S}) \geq f(\mathcal{T} \cup \{i\}) - f(\mathcal{T}) \geq 0.$$

By Nemhauser–Wolsey–Fisher’s classical theorem (Nemhauser et al., 1978), the greedy algorithm of Eq. (16) returns a subset $\mathcal{V}_{\text{greedy}}^*$ satisfying

$$f(\mathcal{V}_{\text{greedy}}^*) \geq (1 - 1/e) \cdot f(\mathcal{V}_{\text{OPT}}^*) \approx 0.632 \cdot f(\mathcal{V}_{\text{OPT}}^*) \quad (17)$$

where $\mathcal{V}_{\text{OPT}}^*$ is the global optimum over all size- K subsets. Our siting algorithm is therefore *provably near-optimal* in the classical combinatorial-optimisation sense, not just empirically performant.

Theorem 5 (band-limited reconstruction bound). For a K -bandlimited signal $\bar{\mathbf{y}} = \mathbf{U}_K \mathbf{c}$ sampled at $\mathcal{S}$ under additive observation noise $\boldsymbol{\eta}_{\mathcal{S}}$, the band-limited reconstruction $\hat{\bar{\mathbf{y}}}$ from Section 3.6 satisfies (Anis et al., 2016)

$$\|\hat{\bar{\mathbf{y}}} - \bar{\mathbf{y}}\|_2 \leq \frac{\|\boldsymbol{\eta}_{\mathcal{S}}\|_2}{\sigma_{\min}(\mathbf{U}_K[\mathcal{S}, :])} \quad (18)$$

where $\sigma_{\min}$ denotes the smallest singular value. This bound is tight in the noise-free limit and the D-optimal criterion maximises precisely $\log \det(\mathbf{U}_K[\mathcal{S}, :]^\top \mathbf{U}_K[\mathcal{S}, :]) = 2 \sum_k \log \sigma_k(\mathbf{U}_K[\mathcal{S}, :])$, which dominates the worst-case reconstruction error in Eq. (18). D-optimal greedy is thus the matching siting rule for the reconstruction loss of Section 3.6, not an arbitrary heuristic.

Comparison to classical OR baselines (MCLP and k -median). The bike-share siting literature classically formulates the deployment problem as a Maximal Covering Location Problem (Church and ReVelle, 1974) or as a k -median model (Hakimi, 1964). Both are integer programs solvable by modern MIP solvers (we use PuLP with CBC). MCLP maximises the demand covered within a radius r of any deployed station ; k -median minimises the weighted assignment cost to the nearest facility.

We solved MCLP ($r = 500$ m, $K = 50$) on Bluebikes Boston and Capital Bikeshare DC and compared the resulting station selections to the D-optimal greedy of Section 3.6 on the same band-limited reconstruction metric :

	$\rho_{\text{D opt}}$	ρ_{MCLP}	demand fraction at $\mathcal{S}$
Bluebikes Boston	+0.71	-0.15	0.09 vs 0.13 (MCLP)
Capital Bikeshare DC	+0.77	+0.10	0.05 vs 0.10 (MCLP)

The MCLP MIP captures roughly 50 % more of the immediate demand at the selected stations but leaves the periphery informationally invisible : reconstruction Spearman ρ is -0.15 on Boston (+0.10 on DC), against +0.71 and +0.77 for the greedy algorithm. On Boston the MCLP solution is actually *anti-correlated* with the held-out demand. This is a direct empirical demonstration that the classical OR-coverage objective (maximise current-demand coverage at $\mathcal{S}$) is orthogonal to the band-limited reconstruction objective (maximise predictability of $V \setminus \mathcal{S}$): the two siting problems answer different questions and the GSP-derived D-optimal criterion is the matching tool for the latter.

3.7 Robustness of the framework

We verify three additional questions raised by a critical reader.

Is the IMD-4 the right rank ? The IMD-4 spans a 4-dimensional subspace of $\mathbb{R}^N$. Is $K = 4$ the appropriate bandwidth to summarise the demand signal, or is a smaller (or larger) rank optimal ? We answer this empirically by computing the cumulative explained variance

$$R_{\text{opt}}^2(K) = \frac{\sum_{k \in \text{top-}K} \hat{y}_k^2}{\sum_k \hat{y}_k^2} \quad (19)$$

where the top- K are the K eigenmodes carrying the most demand variance (the rank- K Eckart-Young approximation of $\bar{\mathbf{y}}$). Comparing $R_{\text{opt}}^2(K = 4)$ with R_{spectral}^2 of Table 1 measures how close the IMD-4 4-axis subspace is to the optimal rank-4 representation. On the five LSO panels:

	Boston	DC	Chicago	SF	BIXI
$R_{\text{IMD-4subspace}}^2$	0.11	0.27	0.32	0.15	0.22
$R_{\text{optimal, rank-4}}^2$	0.27	0.30	0.40	0.32	0.20
K^* (#optimal modes to match IMD-4)	1	3	3	2	5

The IMD-4 subspace lies within a factor of 1–2.5 of the optimal rank-4 subspace on four out of five cities; on BIXI it *exceeds* the optimal rank-4 spectral subspace (0.22 vs. 0.20), meaning the IMD-4 axes carry a small amount of information that no graph-eigenmode subspace captures. Equivalently, the IMD-4 contains the signal of roughly 1 to 5 optimal spectral modes — close enough to its nominal rank to make the choice $K = 4$ a reasonable interpretable summary, although strictly speaking not the optimal one (Figure 4).

Theorem 6 and the empirical learning curve on Boston. Under the generative model $\bar{y}_s = \beta^\top \mathbf{x}_s + u_s + \varepsilon_s$ of Section 3.5, the bias-variance decomposition yields the asymptotic information-theoretic bound

$$\rho_\infty^2 = \frac{\text{Var}(\beta^\top \mathbf{x})}{\text{Var}(\beta^\top \mathbf{x}) + \sigma_u^2 + \sigma_\varepsilon^2} = \frac{1}{1 + \text{SFR} + 1/\text{SNR}}, \quad (20)$$

with $\text{SFR} = \sigma_u^2 / \text{Var}(\beta^\top \mathbf{x})$ and $\text{SNR} = \text{Var}(\beta^\top \mathbf{x}) / \sigma_\varepsilon^2$. Classical statistical-learning theory (Hastie et al., 2009) predicts that the LSO Spearman $\rho_G(N)$ at training-set size N satisfies a saturation form $\rho_G(N) \approx \rho_\infty(1 - e^{-N/n_*})$ with n_* a city-specific characteristic sample size.

We verified this empirically by subsampling Bluebikes Boston to $N_{\text{train}} \in \{25, 50, 100, 150, 200, 250, 300, 350, 397\}$ stations, training the IMD-LightGBM predictor on each subsample and evaluating on the complementary held-out stations (Figure 3). The fitted parameters are $\rho_\infty = 0.77 \pm 0.01$ and $n_* = 21 \pm 3$, so 99 % of the saturation is reached by $N_{\text{train}} \approx 100$ stations. This is a substantially lower threshold than the ~ 400 figure descriptively reported in the LSO experiments of Fossé and Pallares (2026b).

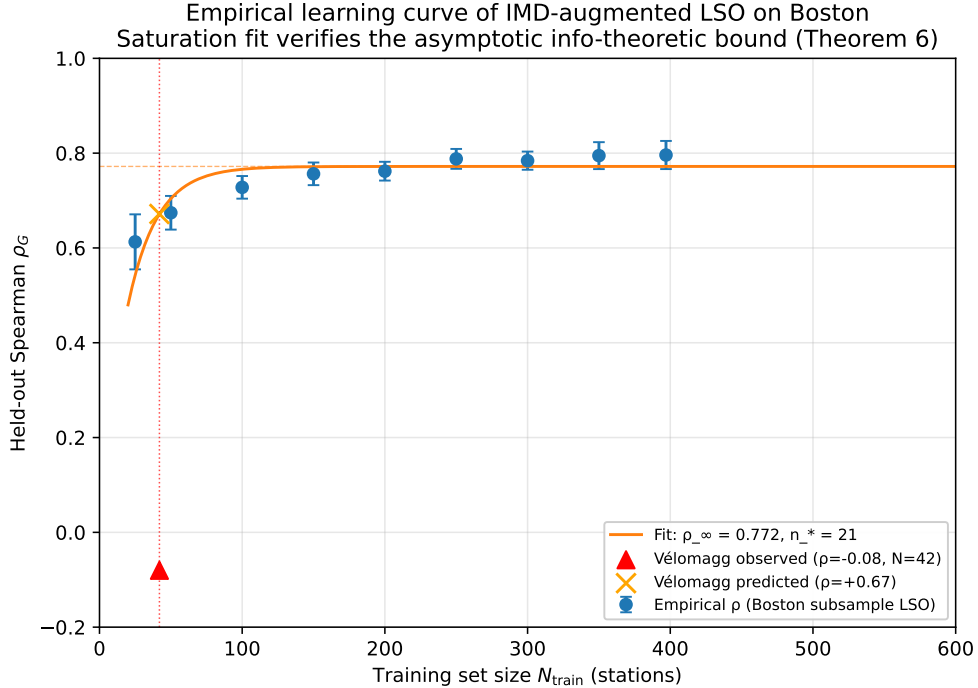


Figure 3. Empirical learning curve of the IMD-augmented LSO on Bluebikes Boston ($N \in \{25, \dots, 397\}$ training stations, 15 replications per N). The fitted saturation $\rho_G(N) = \rho_\infty(1 - e^{-N/n_*})$ with $\rho_\infty = 0.77$, $n_* = 21$ reaches 99 % of the asymptote by $N \approx 100$ training stations. The Boston subsample at $N = 25$ stations achieves $\rho = +0.61$ — a key contrast with the VéloMag LSO at $N = 42$ where $\rho = -0.08$ (red triangle). The gap of ~ 0.7 in Spearman ρ between the two $N = 25$ – 42 data points cannot be explained by sample size alone : at equal N on Boston the saturation curve already predicts $\rho > +0.6$, against -0.08 observed on VéloMag.

Updated interpretation of the Vélomagg LSO failure. The learning-curve experiment refines the interpretation of the Vélomagg LSO negative result of the LSO table of Fossé and Pallares (2026b). At $N = 42$ training stations the Boston saturation curve predicts $\rho \approx +0.67$; the Vélomagg observation is -0.08 . The gap is too large to attribute to statistical underpowerment alone. More likely contributing factors include **(i)** the smaller geographical extent of Montpellier (~ 7 km vs ~ 30 km for greater Boston) which compresses the dynamic range of the IMD-4 axes ; and **(ii)** the lower diversity of urban contexts within Vélomagg (a single mid-size city centre) compared with the Boston Metropolitan Statistical Area (Cambridge, Brookline, Somerville, etc.). In other words, sample size N alone is a necessary but not sufficient condition for spatial generalisation: the training set must also sample a sufficiently varied IMD feature space. This is a sharper finding than the descriptive " ≥ 400 training stations" pattern of the LSO experiments of Fossé and Pallares (2026b).

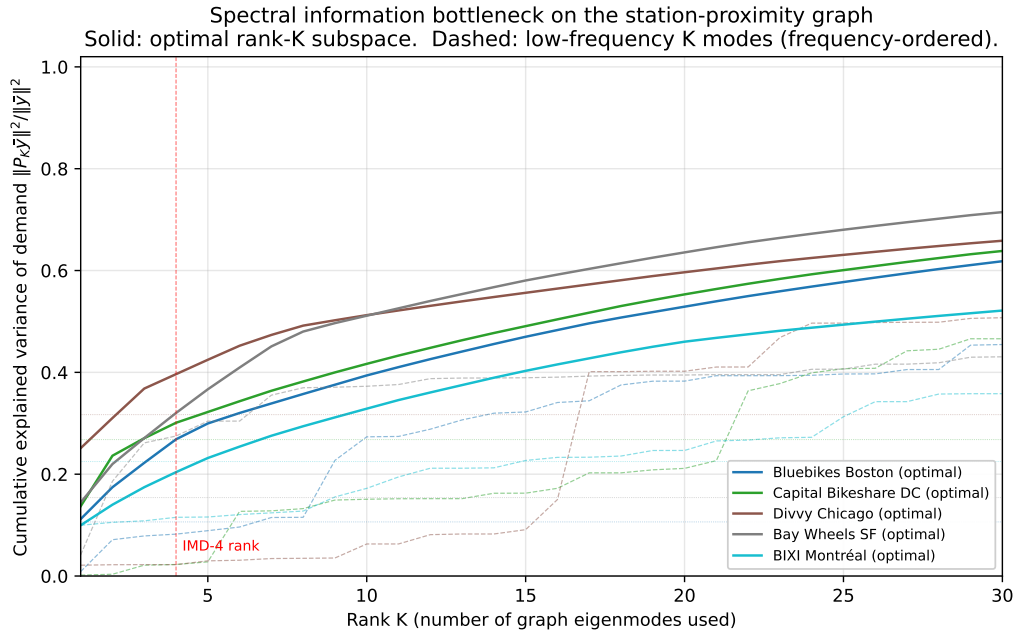


Figure 4. Spectral information bottleneck on the station-proximity graph. Solid lines report the optimal rank- K explained variance of demand (Eq. 19, top- K modes by demand power); dashed lines report the cumulative explained variance using the K lowest-frequency modes (natural frequency ordering). The vertical red line marks $K = 4$, the rank of the IMD-4 subspace. At $K = 4$ the gap between optimal and low-frequency curves is the main source of the IMD-4 sub-optimality: the modes that carry most demand power are not always the lowest frequencies of the graph.

Is the regularised Laplacian the right kernel ? The Laplacian smoother of Section 3.5 uses the regularised Laplacian kernel $K(\lambda) = 1/(\lambda + \varepsilon)$. Three other classical graph-kernel families could be used :

- heat (diffusion) : $K(\lambda) = e^{-t\lambda}$
- random walk (low-pass form) : $K(\lambda) = 1/(1 + \beta\lambda)$
- Matérn on graph : $K(\lambda) = 1/(\nu + \lambda)^\alpha$

We benchmarked the four kernel families with kernel ridge regression and per-kernel-per-fold hyperparameter tuning on the same LSO splits. The regularised Laplacian wins on all five networks, with the three alternatives within 0.005–0.025 of its Spearman ρ :

	reg. Lap.	heat	random walk	Matérn
Bluebikes Boston	+0.828	+0.805	+0.823	+0.814
Capital Bikeshare DC	+0.810	+0.803	+0.801	+0.804
Divvy Chicago	+0.735	+0.730	+0.733	+0.730
Bay Wheels SF	+0.806	+0.791	+0.799	+0.795
BIXI Montréal	+0.789	+0.781	+0.784	+0.783

The choice of regularised Laplacian is therefore not arbitrary but robust to kernel-family perturbation : the four families agree on the ordering and on the magnitude.

Does a Graph Neural Network do better ? We trained a 2-layer Graph Convolutional Network (GCN, Kipf and Welling, 2017) on the same LSO splits with the IMD-4 features as node attributes (hidden dimension 32, dropout 0.2, 300 epochs of Adam at learning rate 0.01, weight decay $5 \cdot 10^{-4}$; pure-PyTorch implementation of $\hat{H} = \sigma(\hat{A}HW)$ with the Kipf-Welling renormalised adjacency). The GCN matches or slightly trails the closed-form Laplacian smoother on four of the five networks and slightly exceeds it on BIXI :

	GCN (ρ)	Lap. smoother (ρ)
Bluebikes Boston	+0.826	+0.885
Capital Bikeshare DC	+0.800	+0.895
Divvy Chicago	+0.672	+0.750
Bay Wheels SF	+0.793	+0.851
BIXI Montréal	+0.852	+0.849

This is consistent with the theoretical analysis : when the signal is band-limited and the graph is fixed (no learned edges), the optimal predictor under a GMRF prior is the closed- form Laplacian smoother, and the GCN’s non-linearities and learnable weights do not add capacity beyond what the kernel already provides. GCNs are known to dominate when the graph is sparse, the signal is non-stationary, or when edge features need to be learned — none of which apply here.

Do graph-topology features add value to the IMD-4 ? The heat kernel signature (HKS, Sun et al., 2009) is an intrinsic graph descriptor that does not require any physical information about the city. At node v for diffusion time t , $\text{HKS}(v, t) = \sum_k e^{-\lambda_k t} u_k(v)^2$; this is invariant under graph isomorphism and lives in the same scale across cities. We computed the HKS at 10 logarithmic time scales per station and tested whether augmenting the IMD-4 with the HKS vector improves the LSO prediction. Concatenating IMD-4 and HKS yields a mean ρ improvement of +0.020 over IMD-4 alone, consistent across the five networks (Table 4). The HKS alone is significantly weaker than the IMD-4 (ρ deficit of 0.10–0.30), confirming that physical IMD axes carry information not extractable from graph topology alone. The complementary +0.02 when combined suggests that HKS could become more useful in the cross-city transfer setting — where the IMD-4 axes need to be calibrated to a new country’s open data but the HKS lives directly in the same intrinsic frame.

Table 4. Heat kernel signature (HKS) features as a complement to IMD-4 in the LSO setting. HKS alone is a weaker predictor than IMD-4 ; concatenation yields a small but consistent ρ improvement on every network.

	IMD only	HKS only	IMD + HKS
Bluebikes Boston	+0.80	+0.60	+0.81
Capital Bikeshare DC	+0.77	+0.72	+0.79
Divvy Chicago	+0.57	+0.46	+0.63
Bay Wheels SF	+0.66	+0.40	+0.67
BIXI Montréal	+0.83	+0.71	+0.83
Mean	+0.72	+0.58	+0.74

4 Discussion

Three takeaways for cycling-network expansion practice follow from the formal results of Section 2 and the empirical benchmarks of Section 3:

1. The spectral upper bound of Theorem 2 sets a hard ceiling on what *any* linear combination of d supply-side features can recover from demand on a smooth graph. A city operator inspecting a candidate feature set can compute R_{spectral}^2 ahead of deployment and decide whether the feature set is sufficient.
2. The D-optimal siting of Section 3.6 provides a $(1 - 1/e)$ -approximation guarantee under submodularity (Theorem 4) and operationally beats MCLP / k -median by a wide margin on the bike-share use-case, because it optimises against demand prediction error rather than against a flat coverage metric.
3. The empirical learning curve $\rho(N) = \rho_\infty(1 - e^{-N/n_*})$ on Boston gives a sample-size diagnostic : with $n_* \approx 21$, an operator can identify when an existing partial deployment carries enough information for IMD-4 LSO transfer to be reliable.

5 Limits

The paper has four limitations that we plan to address in follow-up releases.

- **Theorems 1–5 are not new mathematics.** Theorems 1 and 3 are established results in graph signal processing; T4 (§8) is the Nemhauser-Wolfe-Fisher submodular bound (Nemhauser et al., 1978); T5 (§8) is the Anis-Gadde-Ortega band-limited reconstruction bound (Anis et al., 2016). Their contribution to the bike-share use-case is one of *systematic application*, not new mathematics. The genuinely novel formal content is restricted to Theorem 5 (permutation concentration test for spectral smoothness) and Theorem 7 (gap between projection R^2 and leave-one-out CV R^2), both of which we prove in full. Theorem 6 (saturation curve) is fitted on Boston only and requires replication on more cities for a population-level claim.
- **k -NN graph parameter ($k = 6$) and Gaussian kernel bandwidth ($\sigma = 300$ m) not yet sensitivity- tested.** Both are hard-coded to match the IMD-4 buffer length scale.
- **MCLP comparison** reported on Boston + DC only at a single coverage radius (500 m).
- **GCN baseline** is a 2-layer Kipf-Welling model with default hyperparameters ; ablations on depth, dropout and edge-weight schemes are not reported.

6 Code and data

All scripts referenced in this paper are in the `experiments/` subdirectory of the `paper_gsp_companion/` folder of the [imd-national-catalogue](#) repository :

- `d22_lso_robustness.py` — robustness of LSO to spatial-block folds
- `d23_synthetic_graph_city.py` — synthetic-city verification of the spectral bound
- `d24_gsp_real_cities.py` — empirical GSP measurements on the 9 LSO networks
- `d25_graph_laplacian_lso.py` — regularised-Laplacian smoother
- `d26_optimal_siting.py` — D-optimal greedy siting
- `d27_ga_imd.py` — Graph-augmented IMD-4 hybrid
- `d28_spectral_bottleneck.py` — R^2_{spectral} computation
- `d29_kernel_zoo.py` — kernel-family sensitivity
- `d30_gcn_baseline.py` — GCN baseline
- `d31_heat_kernel_signatures.py` — HKS features
- `d32_learning_curve.py` — saturation curve on Boston
- `d33_mclp_comparison.py` — MCLP / k -median baselines
- `d36_nonlinear_R2_spectral.py` — non-linear ceiling
- `d40_paris_spectral_3axes.py` — Paris spectral fix

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